

Stage 5 Delta Diagnostic Report

Price-Implied Delta and a Supervised Delta-Head Baseline

Codex-assisted surrogate modelling review

June 21, 2026

Abstract

This report reviews Stage 5 of the compressed roadmap. It compares Delta implied by retrained Stage 3 price/premium surrogates against a supervised Delta-head baseline trained only on approved finite-difference Delta diagnostic labels. No Gamma head is trained, Gamma is not used as a target, and the existing v1 dataset is read-only. The Stage 5 review decision is `READY_FOR_INTEGRATED_WORKFLOW_STAGE`.

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1 Purpose of Stage 5

Stage 5 asks whether the existing price/premium surrogates give coherent Delta behavior, and whether a separate supervised Delta head is needed. Delta is more delicate than price because it is a derivative diagnostic of the value surface, especially near payoff kinks, maturity rows, and exercise-boundary regions. The numerical setting follows the finite-difference American-option framework used in the solver validation reports (Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Link to Stage 3 and Stage 4

Stage 3 accepted the positive-premium price surrogate for boundary diagnostics. Stage 4 showed that boundary behavior needed a separate boundary-focused model rather than relying only on threshold extraction from price/premium predictions. Stage 5 applies the same caution to Delta diagnostics.

3 Delta Target Construction and Mask Policy

The target `y_delta` is the finite-difference Delta stored in the `v1` dataset. It is not an analytical Greek and is not a production risk label. Primary training and evaluation use `delta_allowed_mask` and finite `y_delta` rows, giving 1,237,009 allowed rows. The strict-interior mask is the main reliable diagnostic region; boundary-near, payoff-kink-near, and maturity rows are reported as difficult diagnostic slices.

4 Price-Implied Delta Method

The direct-value and positive-premium Stage 3 MLPs are retrained deterministically in memory. Delta is estimated by central finite difference in log moneyness with step 10^{-3} :

$$\hat{\Delta} = \frac{\hat{V}(\log(S/K) + h) - \hat{V}(\log(S/K) - h)}{\exp(\log(S/K) + h) - \exp(\log(S/K) - h)}.$$

For the premium model, payoff is recomputed analytically at the perturbed moneyness and value is reconstructed as payoff plus predicted premium.

5 Supervised Delta-Head Method

The supervised Delta head uses the seven `v1` features and trains only on train-split rows satisfying `delta_allowed_mask`. Predictions are clipped to the basic diagnostic bounds: call Delta in $[0, 1]$ and put Delta in $[-1, 0]$. This is a diagnostic baseline, not production Greek infrastructure.

6 Metrics and Evaluation Protocol

Metrics include Delta MAE, RMSE, maximum absolute error, stable relative MAE, bounds violation rate, and sign violation rate. Bounds and sign rates use tolerance 10^{-4} . Review readiness requires validation and test RMSE at most 0.12, stress-holdout RMSE at most 0.18, and near-zero bounds/sign violations for at least one method.

7 Results by Split

Method	Split	Rows	RMSE	MAE
direct_value_implied_delta	validation	78,983	0.121573	0.092591
direct_value_implied_delta	test	193,872	0.168430	0.131718
direct_value_implied_delta	stress_holdout	103,124	0.125000	0.101264
positive_premium_implied_delta	validation	78,983	0.030056	0.017500
positive_premium_implied_delta	test	193,872	0.057658	0.038061
positive_premium_implied_delta	stress_holdout	103,124	0.050654	0.034781
supervised_delta_head	validation	78,983	0.038170	0.024185
supervised_delta_head	test	193,872	0.057626	0.040306
supervised_delta_head	stress_holdout	103,124	0.048061	0.033768

The Stage 5 gate is satisfied by `supervised_delta_head`. The price-implied Delta methods remain useful diagnostics, but they do not replace the supervised Delta head under the combined RMSE and bounds/sign checks. This supports using a separate Delta-focused diagnostic model rather than assuming price or premium accuracy is enough for Greek behavior.

8 Results by Option Family

Method	Option	RMSE	MAE
direct_value_implied_delta	put	0.108341	0.084589
direct_value_implied_delta	call	0.144553	0.110942
positive_premium_implied_delta	put	0.039833	0.024262
positive_premium_implied_delta	call	0.044374	0.027971
supervised_delta_head	put	0.040893	0.025470
supervised_delta_head	call	0.047333	0.031011

9 Results by Diagnostic Region

Method	Split	Region	Rows	RMSE
direct_value_implied_delta	test	all_allowed	193,872	0.168430
direct_value_implied_delta	test	strict_interior	193,872	0.168430
direct_value_implied_delta	test	boundary_near	2,219	0.149351
direct_value_implied_delta	test	payoff_kink_near	26,015	0.157259
direct_value_implied_delta	test	maturity_row	1,849	0.220151
direct_value_implied_delta	stress_holdout	all_allowed	103,124	0.125000
direct_value_implied_delta	stress_holdout	strict_interior	103,124	0.125000
direct_value_implied_delta	stress_holdout	boundary_near	6,341	0.142873
direct_value_implied_delta	stress_holdout	payoff_kink_near	14,520	0.115300
direct_value_implied_delta	stress_holdout	maturity_row	1,032	0.221023
positive_premium_implied_delta	test	all_allowed	193,872	0.057658
positive_premium_implied_delta	test	strict_interior	193,872	0.057658

Method	Split	Region	Rows	RMSE
positive_premium_implied_delta	delta	boundary_near	2,219	0.046360
positive_premium_implied_delta	delta	payoff_kink_near	26,015	0.170213
positive_premium_implied_delta	delta	maturity_row	1,849	0.051932
positive_premium_implied_delta	delta	all_allowed	103,124	0.050654
positive_premium_implied_delta	delta	strict_interior	103,124	0.050654
positive_premium_implied_delta	delta	boundary_near	6,341	0.018641
positive_premium_implied_delta	delta	payoff_kink_near	14,520	0.145601
positive_premium_implied_delta	delta	maturity_row	1,032	0.061880
supervised_delta_head	test	all_allowed	193,872	0.057626
supervised_delta_head	test	strict_interior	193,872	0.057626
supervised_delta_head	test	boundary_near	2,219	0.121994
supervised_delta_head	test	payoff_kink_near	26,015	0.101857
supervised_delta_head	test	maturity_row	1,849	0.169265
supervised_delta_head	stress_holdout	all_allowed	103,124	0.048061
supervised_delta_head	stress_holdout	strict_interior	103,124	0.048061
supervised_delta_head	stress_holdout	boundary_near	6,341	0.077442
supervised_delta_head	stress_holdout	payoff_kink_near	14,520	0.074468
supervised_delta_head	stress_holdout	maturity_row	1,032	0.173427

10 Bounds and Sign Diagnostics

Method	Split	Bounds rate	Sign rate	Review
direct_value_implied_delta	validation	0.383652	0.220693	REVIEW
direct_value_implied_delta	test	0.217705	0.172304	REVIEW
direct_value_implied_delta	stress_holdout	0.204579	0.189170	REVIEW
positive_premium_implied_delta	validation	0	0	PASS
positive_premium_implied_delta	test	0.001259	0	REVIEW
positive_premium_implied_delta	stress_holdout	0	0	PASS
supervised_delta_head	validation	0	0	PASS
supervised_delta_head	test	0	0	PASS
supervised_delta_head	stress_holdout	0	0	PASS

11 Figures

12 Limitations

Delta labels are finite-difference diagnostics from the CN/PSOR dataset. Good price or premium performance does not automatically imply good Delta behavior. Boundary-near, payoff-kink-near, and maturity rows remain delicate. Gamma is deliberately excluded from training and remains blocked from this stage.

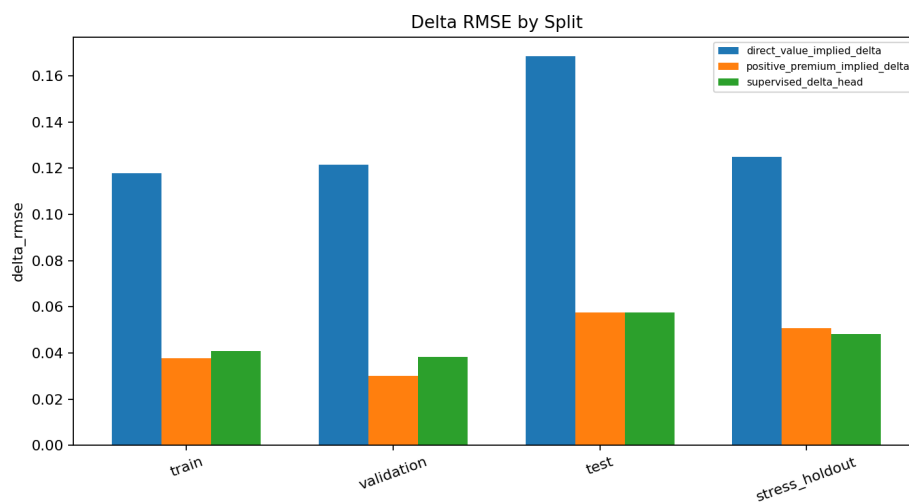


Figure 1: Delta error by split.

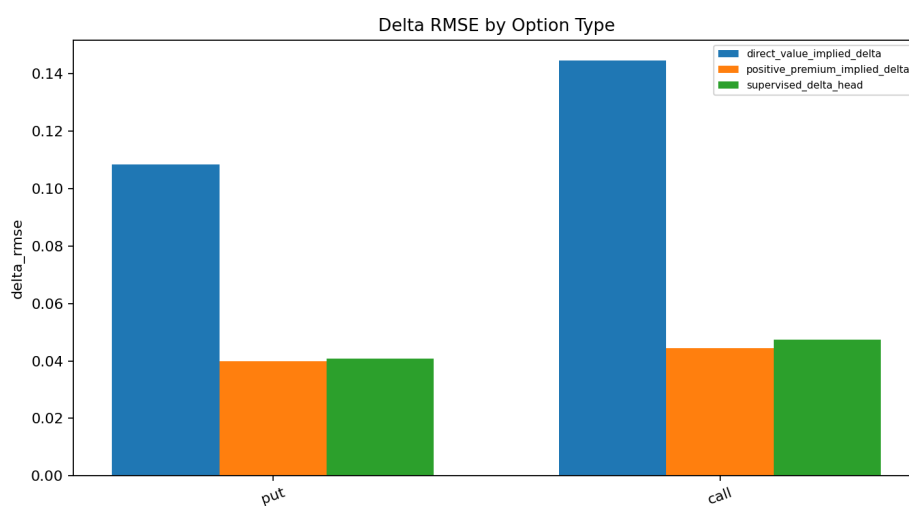


Figure 2: Delta error by option type.

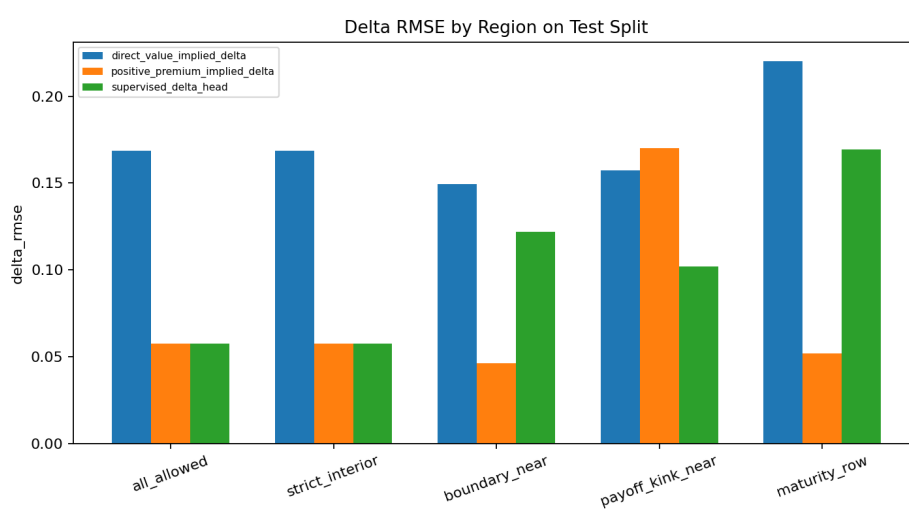


Figure 3: Delta error by region.

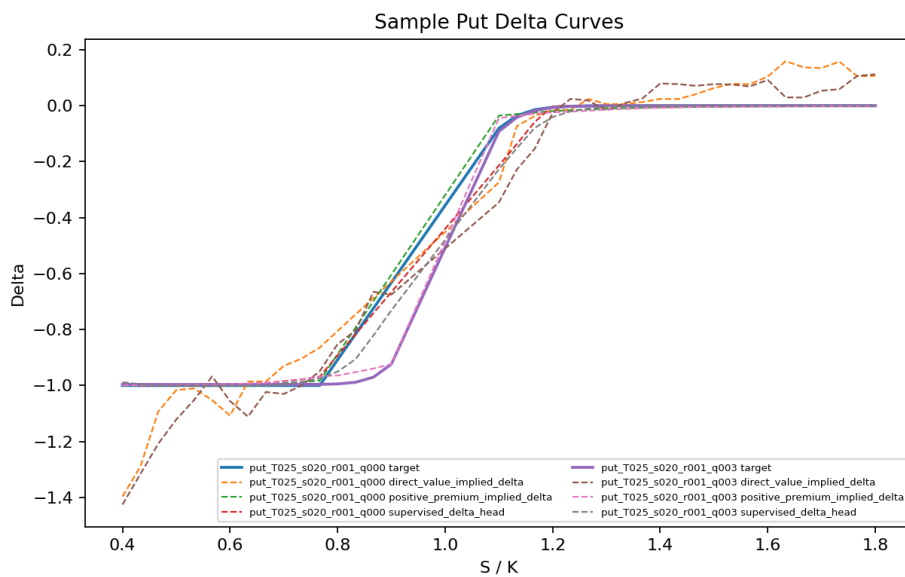


Figure 4: Sample put delta curves.

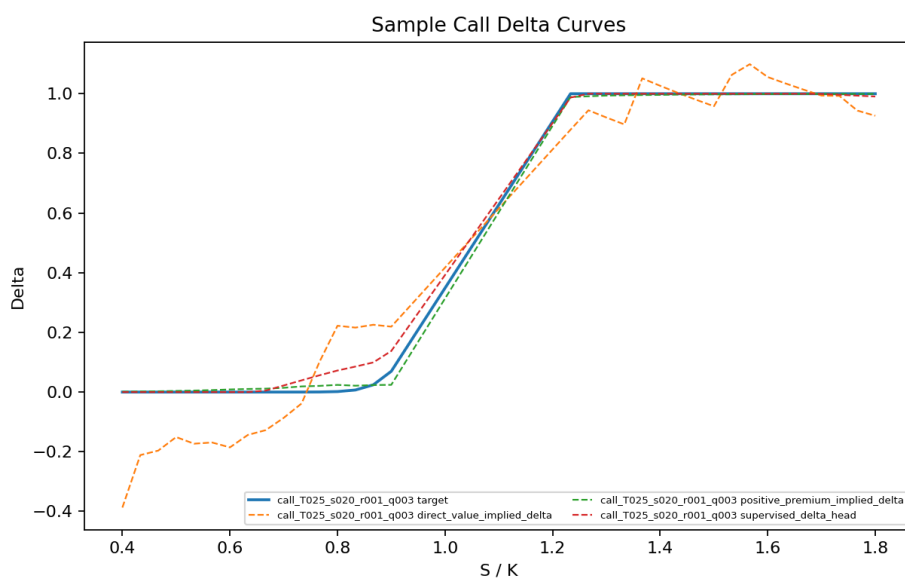


Figure 5: Sample call delta curves.

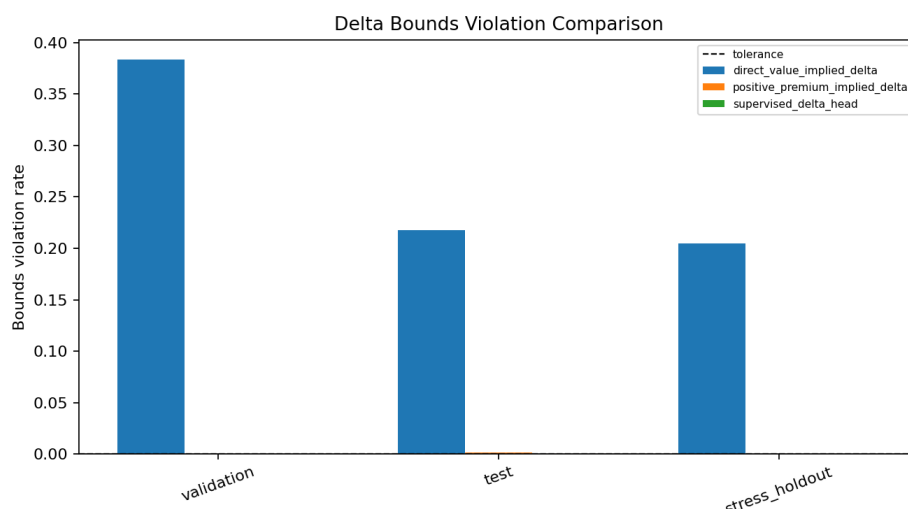


Figure 6: Delta bounds violation comparison.

13 What Stage 5 Supports

Stage 5 supports comparing price-implied Delta diagnostics with a supervised Delta-head baseline under the existing v1 masks and regime-level splits. If the supervised head is the passing method, the evidence supports using a separate Delta-focused diagnostic model.

14 What Stage 5 Does Not Support

This stage does not support production Greek reliability, Gamma-head training, exact analytical Greek claims, final paper claims, solver changes, or v1 dataset regeneration.

15 Recommended Next Stage

If the review decision is `READY_FOR_INTEGRATED_WORKFLOW_STAGE`, the next stage may integrate the price, boundary, and Delta diagnostic components into a workflow review. If the decision is `REVIEW_REQUIRED_BEFORE_INTEGRATED_STAGE`, human review should inspect the Delta failures before adding new components.

References

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.